

Little evidence for meaningful intervention effect heterogeneity in resistance training: Evidence from causal inference and meta-analytic variance comparisons*

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Abstract

Resistance training studies consistently demonstrate substantial variation in observed outcomes between participants. This variation is commonly interpreted as evidence that individuals differ meaningfully in their responses to training interventions. However, observed variation in outcomes does not necessarily imply heterogeneity in the underlying causal effects of an intervention. Rather, observed outcomes reflect multiple sources of variation, including between-participant differences, within-participant variability, measurement error, and potentially true intervention effect heterogeneity. Within the potential outcomes framework, heterogeneity of intervention effects is fundamentally a question about the variance of causal effects. Although directly estimating this variance generally requires specialised study designs such as randomised replicated crossover trials, traditional randomised controlled trials (RCTs) comparing an intervention against an inert control can provide indirect evidence regarding intervention effect heterogeneity through comparisons of outcome variances. Here, I outline the causal framework underpinning variance comparisons, discuss the assumptions required for causal interpretation, and demonstrate how meta-analytic models can synthesise evidence across studies to compare variances while accounting for the ubiquitous relationship between means and variances. Applying these methods to a large meta-analytic dataset of RCTs of resistance training interventions reveals little evidence that interventions meaningfully increase outcome variability beyond that expected from the increase in mean outcomes themselves. Although conclusions necessarily depend upon assumptions regarding the correlation between potential outcomes, these assumptions appear empirically reasonable for resistance training studies in healthy participants. Taken together, current evidence provides little support for the existence of substantial intervention effect heterogeneity in resistance training adaptations. **Keywords:** resistance training, heterogeneity, individual response, causal inference, meta-analysis

Introduction

Variation in responses to resistance training has become a topic of considerable interest within sport and exercise science. Large differences are routinely observed between participants following apparently identical training programmes for the outcomes captured, and these differences are frequently interpreted as evidence that individuals differ meaningfully in their responsiveness to training. Such interpretations underpin discussions of “responders” and “non-responders”, personalised exercise prescription, and the search for biological determinants of intervention responsiveness. However, most people working on this topic confuse the observed variation in *outcomes* with there necessarily being heterogeneity in the *underlying effect of an intervention*. This typically has involved merely considering the distribution of outcomes observed in the intervention arms of trials which is insufficient for determining whether such variance is due to the effects of

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the intervention themselves varying between individuals. But, careful consideration of the underlying causal structure of the question of intervention effect heterogeneity allows us to better evaluate whether there is or is not evidence suggesting it is present.

Partitioning of variance components of observed outcomes such that we can draw inferences regarding the presence (or not) of heterogeneity in the underlying causal effect of treatment is incredibly difficult. Typically, study designs such as those used by Robinson et al. (2025) (i.e., randomised replicated crossovers) are required. But, in the context of a traditional randomised controlled trial (RCT) where the control can be reasonably assumed to be inert (i.e., has no effect itself), such as a non-training control in studies of healthy untrained^a participants, we can obtain some insight into whether interindividual heterogeneity in intervention effects may be present.

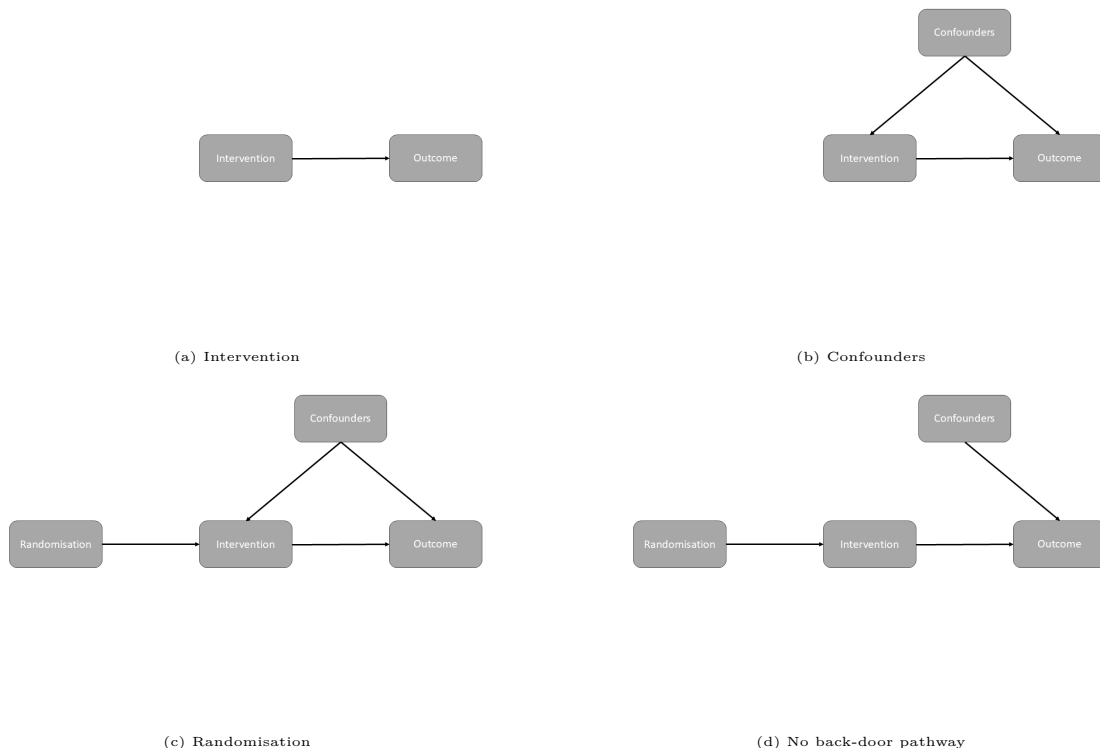


Figure 1: Logic of randomised controlled trial design shown via directed acyclic graphs for estimating average intervention effects. In (a) we want to estimate the causal effect of an intervention upon some outcome, but as in (b) there are often confounders that influence whether the intervention is participated in and the outcome. As such, we randomise participants to receive the intervention (c) and now this means that we can obtain an unbiased estimate of the intervention effect because the only thing that causes participation in the intervention is randomisation (d) i.e., there is no “back-door” causal pathway between the outcome and the intervention.

The logic of the RCT design (Figure 2) allows us to obtain an unbiased estimate of the clearly defined estimand that is the average intervention effect (Equation 1) through comparison of the marginal distributions of outcomes between the conditions e.g., intervention vs control (Equation 2).

$$\begin{aligned} &\text{Estimand: Average intervention effect} \\ &D = Y_{\text{Int}} - Y_{\text{Con}} \end{aligned} \tag{1}$$

^aUntrained participants given in a trained population we might expect *a priori* that a detraining effect may occur after cessation of their training. In the case of trained participants, as others have argued (**ADD REF**), continuation of existing training may be a better control condition and indeed especially so when participants are already sufficiently trained (i.e., having trained consistently and using known effective interventions for at least one full year **ADD REFS**) such that they are no longer experiencing meaningful intervention effects anyway.

Estimate: marginal distributions of outcomes between the conditions

$$\bar{D} = \bar{Y}_{Int} - \bar{Y}_{Con} \quad (2)$$

This estimand, along with its standard deviation (Equation 3), are well defined under the potential outcomes framework (Caldwell et al., 2025).

Standard deviation of intervention effect

$$\sigma_D^2 = \sigma_{Int}^2 + \sigma_{Con}^2 - 2\sigma_{Int}\sigma_{Con}\rho_{Int:Con} \quad (3)$$

The average causal effect of an intervention in comparison to a control can be easily estimated using a simple general linear model:

$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_1 x_i \quad (4)$$

where y_i is the outcome, x_i is a categorical predictor for the condition i , β_0 reflects the average outcome in the reference condition (the inert control in this case), and the average intervention effect is estimated by β_1 .

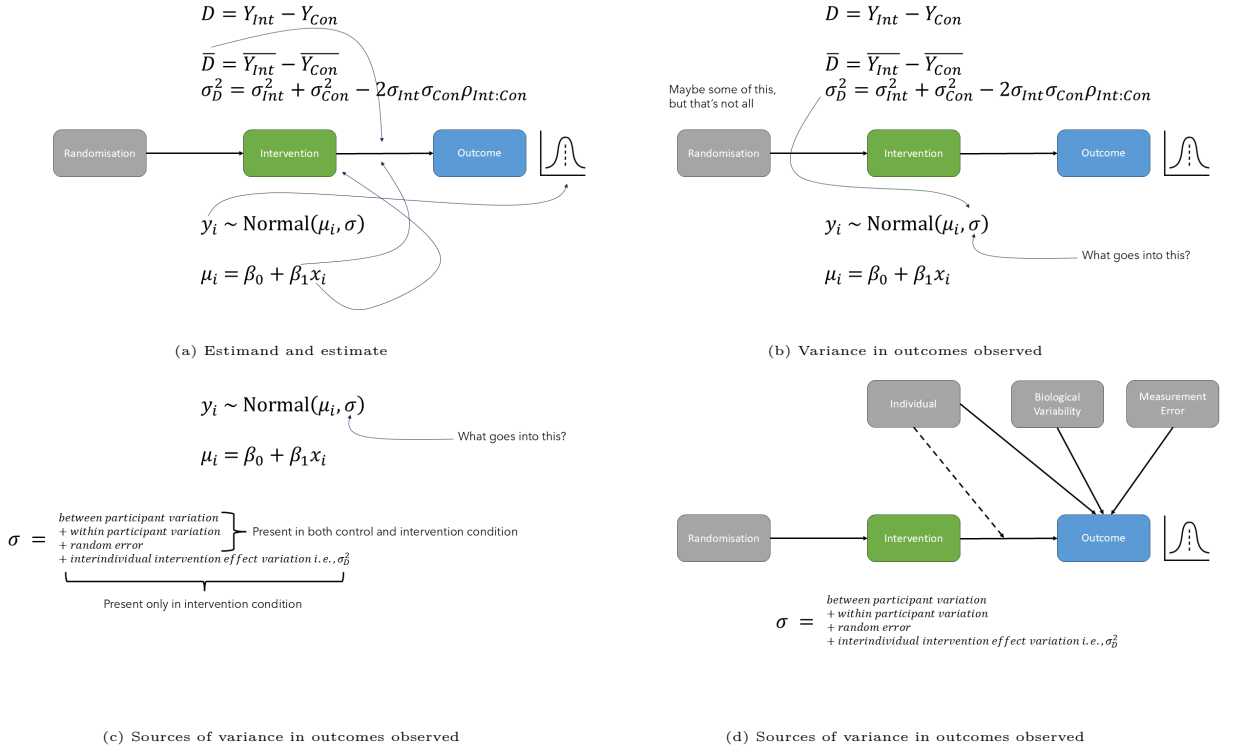


Figure 2: Mapping of estimand and estimate in the randomised controlled trial design for estimating average intervention effects alongside the directed acyclic graph. In (a) we have defined estimand of the average treatment effect, our estimate based on the marginal distributions of interventions and controls (i.e., the causal arrow from intervention to outcome), and the standard deviation of treatment effects in addition to the general linear model estimator used to yield and estimate of this average treatment effect. The components of the general linear model are mapped to their corresponding components in the causal structure including the observed outcomes y_i , the randomly assigned intervention x_i , and the estimated average causal effect of the intervention compared to the control β_1 . In (b), (c), and (d) it shows that some of the variance in observed outcomes i.e., σ , may be due to intervention effect heterogeneity i.e., σ_D^2 (the dashed arrow in (d) depicting interaction of the individual with the intervention), but that there are numerous other sources of variance and that these also differ between the intervention and control conditions.

The question of whether an intervention has introduced variation, that is to say that there is some variation in the intervention effect and not merely a fixed effect amongst participants, is essentially a question of

comparing variances in this type of study design i.e., an RCT. In the context of an RCT with an inert control the variance σ in observed outcomes y_i is due to between participant variation, within participant variation, and other sources of random error in the inert control condition, whereas in the intervention condition only will there potentially be the additional source of variation that is the interindividual variation of the intervention effect itself (i.e., σ_D^2).

For RCTs it is a simple extension of the traditional general linear model to a so-called location-scale model in which variances are modelled directly, for example including the condition as a predictor in the model for this parameter:

$$\begin{aligned} y_i &\sim \text{Normal}(\mu_i, \sigma_i) \\ \mu_i &= \beta_0 + \beta_1 x_i \\ \sigma_i &= \alpha_0 + \alpha_1 x_i \end{aligned} \tag{5}$$

where now α_0 reflects the variance in the control condition and α_1 estimates the effect of the intervention on the variance.

There are also various effect size statistics that can be calculated for the comparison of variances such as the standard deviation of individual responses (SD_{IR}) (Atkinson & Batterham, 2015; Hopkins, 2015), the log ratio of standard deviations ($\ln VR$) and the log coefficient of variation ratio ($\ln CVR$) (Nakagawa et al., 2015; Senior et al., 2020), as well as various other statistical tests for the comparison of variances (Mills et al., 2021). These effect size statistics are particularly useful in the context of meta-analytic approaches for examining the presence (or not) of interindividual intervention effect heterogeneity. However, these effects sizes do come with inherent assumptions that need to be considered when drawing inferences about effect heterogeneity in randomised trials. These include strong assumptions about the correlation between potential outcomes (i.e., the value of the unobservable $\rho_{Int:Con}$) as well as assumptions about the relationship between means and variances (i.e., the SD_{IR} and $\ln VR$ assume the relationship to be zero, whilst the $\ln CVR$ assumes it to be proportional).

The fact that $\rho_{Int:Con}$ is not directly observable has been termed the fundamental problem of causal inference and is why in RCTs we focus upon the average intervention effect as the estimand of interest (because we can only observe a outcomes under a single condition per individual participant and not their potential outcome in the unobserved condition). Caldwell et al. (2025) show that the assumed value of the unobservable $\rho_{Int:Con}$ in fact depends on the ratio between the observed standard deviations in the intervention and control conditions (see <https://osf.io/d59ru/files/7f6sq>).

The mean-variance relationship also is ubiquitous in most natural systems, and certainly in outcomes typical of resistance training studies (Nuzzo et al., 2024; Steele et al., 2023), largely due to errors being multiplicative i.e., increase as the mean increases. As such, given that in the context of a RCT of a resistance training intervention compared to an inert control we typically expect an increase in the mean of outcomes such as strength or hypertrophy (assuming the intervention is on average effective), any variance comparisons should account for this possible impact in order to differentiate this coupling of the parameters from the introduction of variance due to the intervention *qua* intervention. For a single trial, either the location-scale model or use of the $\ln CVR$ is thus preferable.

A further issue is that the typical RCT in resistance training is woefully underpowered to detect what could be considered realistic variance ratios that we may want to detect between controls and interventions. For example, in their simulations Caldwell et al. (2025) show that the typical study of $n=20-30$ participants may have $<25\%$ power to detect these differences and thus the introduction of additional variance by the intervention. Thus, meta-analytic approaches are incredibly value for exploring this question, and with meta-analytic approaches we are also able to go one step farther and directly model the ratio of variances between interventions and controls whilst adjusting for the mean outcomes.

Considering the causal logic of the RCT, and the assumptions that need to be examined when making variance comparisons for the purpose of inference regarding intervention effect heterogeneity, in this paper I re-examine the question of whether there is meaningful resistance training intervention effect heterogeneity.

To do so I take a meta-analytic approach and use the large scale dataset of RCTs examining resistance training interventions compared with inert controls that we have previously presented (Steele et al., 2023).

Methods

All data, code, and materials for the present work are available at: ADD LINKS. All packages used for analysis are cited in the corresponding report generated by the grateful package.

Dataset

In Steele et al. (2023) we utilised the studies previously systematically reviewed by Polito et al. (2021) which were all RCTs of resistance training interventions compared to non-training controls where it could be expected that the control conditions were essentially inert. We re-extracted data from the studies they identified and full details of this dataset can be seen in Steele et al. (2023) in addition to the data being openly available (INSERT LINKS). Here I limit the dataset to only studies of untrained participants (see footnote 1) and explore the both the pooled strength and pooled hypertrophy outcomes.

Analysis

An arm-based approach was taken using a multilevel meta-analytic model of the log transformed standard deviations for each outcome at post-intervention regressed upon the log transformed means for each outcome at post-intervention and the arm from which these variances came (i.e., intervention or control) allowing for comparison of variances between arms whilst adjusting for the population mean-variance relationship. The model was thus:

$$\ln(\hat{\sigma}_{ijk}) = \beta_0 + \tau_{1i} + \tau_{2j} + \tau_{3k} + \beta_1 \ln(\hat{\mu}_{ijk}) + \beta_2 x_{Cond} + \epsilon_{ijk} + m_{ijk} \quad (6)$$

where $\ln(\hat{\sigma}_{ijk})$ is the log transformed standard deviation of outcomes for the k^{th} outcome in the j^{th} arm in the i^{th} study, $\ln(\hat{\mu}_{ijk})$ is the log transformed mean of outcomes with the same subscripts, β_0 is the average log transformed standard deviation in the control arms, β_1 is the slope of the log transformed mean upon the log transformed standard deviation, τ_{1i} , τ_{2j} , and τ_{3k} are the respective normally distributed variances for study, arm, and effect, ϵ_{ijk} and m_{ijk} are similarly the residual variance and sampling errors respectively, finally x_{Cond} is a categorical predictor indicating whether and effect is from a control or intervention arm and β_2 reflects the estimate of the average difference in variances between the control and intervention arms.

As noted variance comparisons in RCTs inherently make assumptions regarding the unobservable correlation between potential outcomes ($\rho_{Int:Con}$). As such, in order to draw justifiable conclusions from the analysis I perform here we should check these assumptions. I calculate this for all the included variance contrasts in the meta-analytic dataset and examine whether the empirical distribution of assumed values are a justifiable assumption to allow us to treat our variance comparisons as indicative of whether there is, or is not, any interindividual intervention effect heterogeneity (see <https://osf.io/cg63r>).

To do so I examine with the same dataset the meta-analytic estimate for $\rho_{Pre:Post}$ in both intervention and control groups. Typically using the marginal estimate $\bar{D} = \bar{Y}_{Post} - \bar{Y}_{Pre}$ is not an unbiased estimated of $D = Y_{Int} - Y_{Con}$ and can be misleading regarding the presence, or not, of interindividual intervention effect heterogeneity². However, in the case of resistance training interventions in healthy participants with strength/hypertrophy outcomes it is in fact a relatively unbiased estimate of D ³. As such, $\rho_{Pre:Post}$ might be a reasonable estimate of $\rho_{Int:Con}$ and so I compare this with the calculated values of assumed $\rho_{Int:Con}$.

Results

and the meta-analytically estimated (using a similar multilevel model structure as noted above) values of $\rho_{Pre:Post}$ are in line with the values of $\rho_{Pre:Post}$ (Strength Outcomes: Intervention groups

$\text{__}(\text{Pre:Post})=0.500$ [95%CI:0.365, 0.614], Control groups $\text{__}(\text{Pre:Post})=0.867$ [95%CI:0.705, 0.943]; Hy-
 pertrophy Outcomes: Intervention groups $\text{__}(\text{Pre:Post})=0.827$ [95%CI:0.762, 0.875], Control groups
 $\text{__}(\text{Pre:Post})=0.965$ [95%CI:0.900, 0.988]).